

YFM FUSION COMPANION

Illustrated User Guide

Yu-Gi-Oh! Forbidden Memories - Windows x64 companion

Manual fusion tools work without an emulator

Optional read-only live and save integration for NTSC-U RetroArch/SwanStation

Independent fan-made utility - no ROM, BIOS, save, or card artwork included

Install and start

Use the release ZIP - GitHub's source archive is for developers

Four steps

- Download the latest yfm_companion_windows_x64 release ZIP from the Releases page.
- Extract the entire ZIP. Never run the program from inside the ZIP preview.
- Run install_dependencies.cmd first. It checks Windows x64 and verifies that the executable and database stayed together.
- Start YFM Fusion Companion.exe. The bottom line should report 722 cards and 25,146 resolved fusion pairs.

What the dependency check means

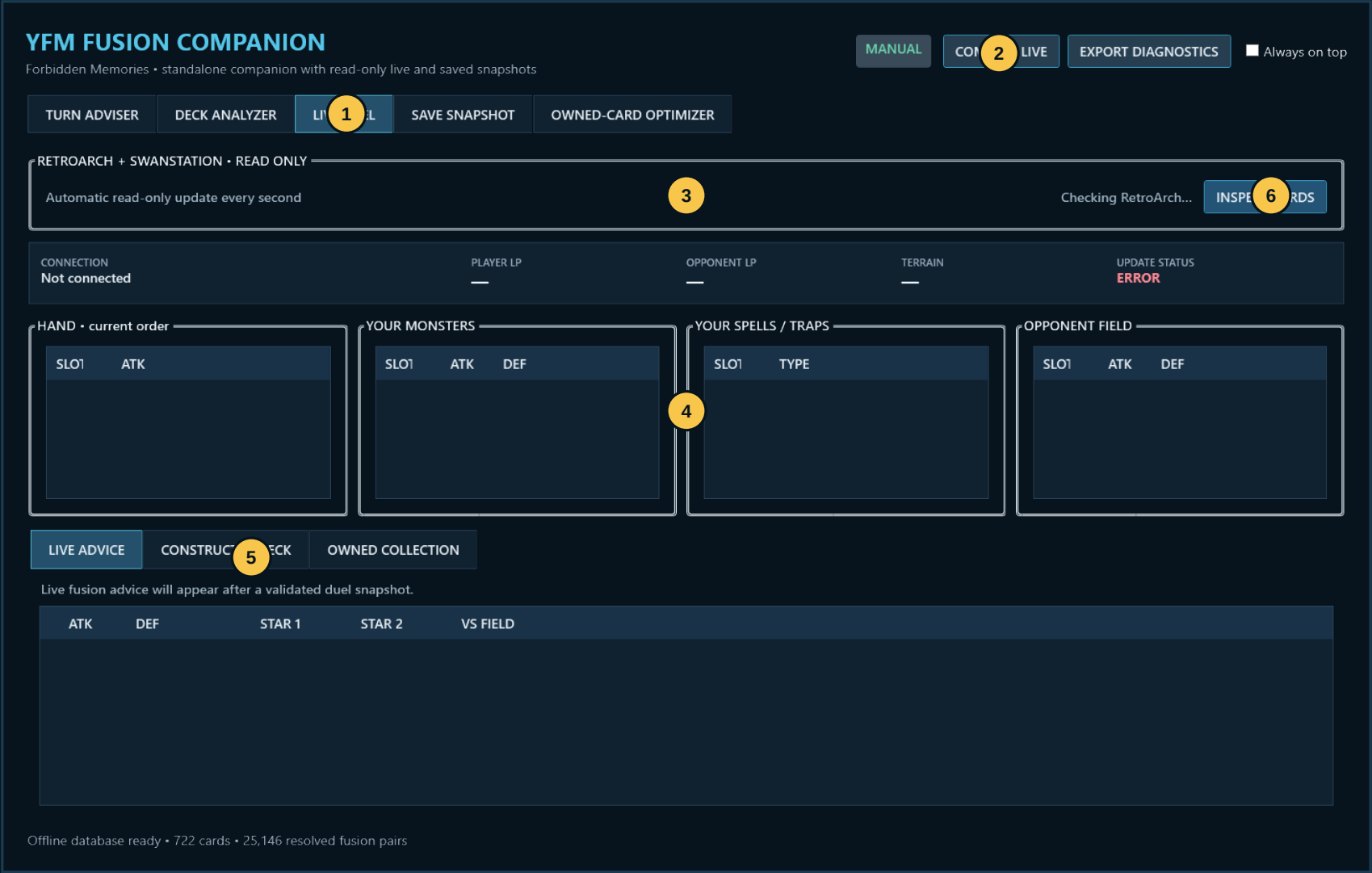
- The portable program is self-contained. Ordinary users do not install .NET.
- RetroArch/SwanStation is optional. It is only needed for Live Duel and automatic save discovery.
- Windows 10 or 11 x64 is required. The release is not code-signed, so SmartScreen may appear.
- Developers can run install_dependencies.ps1 -InstallBuildTools to check or install the .NET 9 SDK.

Optional RetroArch setup

- RetroArch: Settings > Network > Network Commands = ON; Network Command Port = 55355.
- Leave Network RetroPad and stdin Commands OFF. Restart RetroArch once, then launch the NTSC-U game with SwanStation.
- The companion connects to 127.0.0.1 and sends read-only status and memory-read requests. Keep unsolicited inbound UDP 55355 blocked in Windows Firewall.

Interface map

The full window keeps source, display, troubleshooting, and workspace controls visible



- 1

Workspace tabs select the job.
- 2

Source badge and Compact Live are global.
- 3

Read-only connection and summary.
- 4

Hand and player/opponent field tables.
- 5

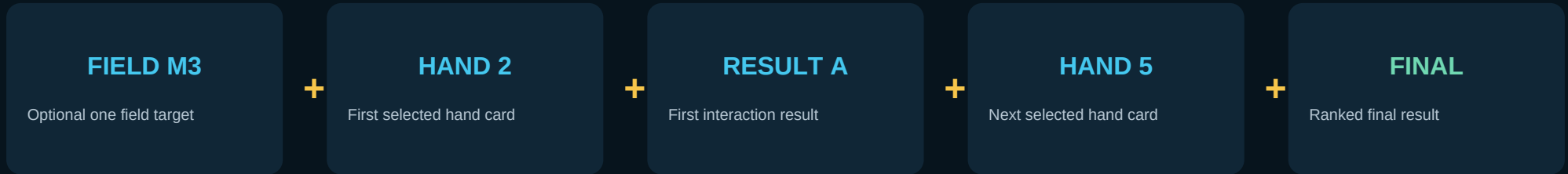
Advice, deck, and collection subtabs.
- 6

Inspector opens only when requested.

Turn Adviser

Enter only cards that actually exist in the current hand and player field

Legal selection order



Inputs and controls

- Hand: up to five ordered cards.
- Monster Field and Spell/Trap Field: up to five optional positions each.
- Analyze Turn ranks legal routes by final ATK, then DEF.
- Tab accepts the top autocomplete result and moves to the next input.
- Empty slots are ignored; duplicate copies remain distinct by slot.

Rules the route obeys

- The field card, when used, interacts with the first selected hand card.
- Only one occupied field position may participate.
- Intermediate results continue through later hand cards in the displayed order.
- Non-fusions that merely discard a material are omitted.
- Compatible equips are terminal; their bonus cannot survive a later monster fusion.
- Documented glitches can be included or excluded explicitly.

Deck Analyzer

Every physical five-card hand is examined for a complete 40-card deck

YFM FUSION COMPANION

Forbidden Memories • standalone companion with read-only live and saved snapshots

MANUAL

COMPACT LIVE

EXPORT DIAGNOSTICS

☐ Always on top

TURN ADVISER

DECK ANALYZER

LIVE DUEL

SAVE SNAPSHOT

OWNED-CARD OPTIMIZER

DECK • up to 40 cards

Deck 1

Deck 2

Deck 3

Deck 4

Deck 5

Deck 6

Deck 7

Deck 8

Deck 9

Deck 10

Deck 11

Deck 12

Deck 13

Deck 14

Deck 15

Deck 16

Deck 17

Deck 18

Deck 19

Deck 20

Deck 21

Deck 22

Deck 23

Deck 24

ANALYZE ALL 5-CARD HANDS

LOAD CURRENT FROM SAVE

CANCEL

CLEAR

ANY FUSION / EQUIP

ATK ≥ 2500

ATK ≥ 2800

ATK ≥ 3000

EXPECTED BEST ATK

DEAD HAND

☒ Include documented glitch fusions

Fill any number of slots. A 40-card deck examines exactly 658,008 physical five-card hands.

FINAL ATK

FINAL DE

HAND CHANC

HANDS

GLITCH

Offline database ready • 722 cards • 25,146 resolved fusion pairs

1

Enter up to 40 cards.

2

Load the validated saved deck in one click.

3

Run, cancel, clear, and glitch controls.

4

Exact probability and expected-ATK metrics.

5

Per-result hand chance and representative route.

YFM Fusion Companion - public Windows x64 guide

Page 5

Live Duel

Validated state refreshes automatically every second

YFM FUSION COMPANION

Forbidden Memories • standalone companion with read-only live and saved snapshots

MANUAL

COMPACT LIVE

EXPORT DIAGNOSTICS

☐ Always on top

TURN ADVISER

DECK ANALYZER

LIVE DUEL

SAVE SNAPSHOT

OWNED-CARD OPTIMIZER

RETROARCH + SWANSTATION • READ ONLY

Automatic read-only update every second

Checking RetroArch...INSPECT CARDS

CONNECTION
Not connected

PLAYER LP

OPPONENT LP

TERRAIN

UPDATE STATUS
ERROR

HAND • current order

SLO1

ATK

3

YOUR MONSTERS

SLO1

ATK

DEF

YOUR SPELLS / TRAPS

SLO1

TYPE

4

OPPONENT FIELD

SLO1

ATK

DEF

LIVE ADVICE

CONSTRUCTED DECK

OWNED COLLECTION

Live fusion advice will appear after a validated duel snapshot.

ATK

DEF

STAR 1

STAR 2

VS FIELD

5

Offline database ready • 722 cards • 25,146 resolved fusion pairs

1

Connection and read-only status.

2

Life Points, terrain, and update health.

3

Current ordered five-card hand.

4

Player and opponent active field positions.

5

Best legal live routes and both guardian-star chains; F#? preserves unknown opponent state.

6

Open the selected-card inspector.

YFM Fusion Companion - public Windows x64 guide

Page 6

Card Inspector

Select a live card, then inspect basic and advanced catalog data.

YFM FUSION COMPANION

Forbidden Memories • standalone companion with read-only live and saved snapshots

MANUAL

COMPACT LIVE

EXPORT DIAGNOSTICS

☐ Always on top

TURN ADVISER

DECK ANALYZER

LIVE DUEL

SAVE SNAPSHOT

OWNED-CARD OPTIMIZER

RETROARCH + SWANSTATION • READ ONLY

Automatic read-only update every second

CONNECTION
Not connected

PLAYER LP

OPPONENT LP

TERRAIN

HAND • current order

SLO1

ATK

YOUR MONSTERS

SLO1

ATK

DEF

YOUR SPELLS / TRAPS

SLO1

TYPE

LIVE ADVICE

CONSTRUCTED DECK

OWNED COLLECTION

Live fusion advice will appear after a validated duel snapshot.

ATK

DEF

STAR 1

STAR 2

VS FIELD

Offline database ready • 722 cards • 25,146 resolved fusion pairs

CARD INSPECTOR

CLOSE

Select a card in the Live Duel tables to inspect its data.

2

3

4

- 1

Select a card in any live table.
- 2

Inspector slides over the right edge.
- 3

Basic type, ATK, and DEF appear first.
- 4

Advanced Data expands catalog relationships.

Save Snapshot

Imports supported memory-card images without modifying them

YFM FUSION COMPANION

Forbidden Memories • standalone companion with read-only live and saved snapshots

MANUAL

COMPACT LIVE

EXPORT DIAGNOSTICS

☐ Always on top

TURN ADVISER

DECK ANALYZER

LIVE DUEL

SAVE SNAPSHOT

OWNED-CARD OPTIMIZER

SWANSTATION MEMORY CARD • READ ONLY

REFRESH SAVED SNAPSHOT

CHOOSE SAVE FILE...

LOAD THIS DECK INTO ANALYZER

LOAD OWNED S INTO OPTIMIZER

Refresh reads the latest Forbidden Memories memory-card save. The load buttons copy that saved deck or collection into the named analysis screen; they never change the game save. Looking for a saved memory-card snapshot...

SOURCEManual entry

FILE—

SAVED—

VALIDATIONNo saved snapshot loaded.

CONSTRUCTED DECK • 40 saved slots

SLOT	#	TYPE	ATK	DEF
------	---	------	-----	-----

COLLECTION • chest + constructed deck

#	CHEST	DECK	TOTAL OWI	IN LIBRA
---	-------	------	-----------	----------

Offline database ready • 722 cards • 25,146 resolved fusion pairs

- 1

Refresh discovery or choose an SRM/MCR file.
- 2

Source, timestamp, and validation details.
- 3

Saved 40-card constructed deck; incomplete decks cannot be loaded into Deck Analyzer.
- 4

Chest, deck, total owned, and Library flag.
- 5

Copy snapshot data into the two analysis tabs.

Campaign and owned-card optimizer

Builds a legal 40-card deck for general play, one duelist, or the final gauntlet

YFM FUSION COMPANION

Forbidden Memories • standalone companion with read-only live and saved snapshots

MANUALCOMPACT LIVEEXPORT DIAGNOSTICS

Always on top

TURN ADVISERDECK ANALYZERLIVE DUELSAVE SNAPSHOTOWNED-CARD OPTIMIZER

STRATEGY • choose the job this deck must do

PROFILE

YOUR FOCUS TYPES • comma separated

Automatic / none

OPPONENT TYPES • comma separated

CAMPAIGN GOAL

TARGET DUELIST

General campaign • i

Choose "One specific opponent" to select a duelist.

Raises Raigeki, broad traps, stall, debuffs, and matchup-relevant removal.

USE SAVED CHIPS

Load a saved snapshot to use its Star Chip budget; the optimizer otherwise uses only the owned quantities below.

ALREADY REDEEMED PASSWORD CARD NAMES • comma separated

Include documented glitch fusions

BUILD OPTIMAL 40-CARD DECKCANCELCLEAR OWNEDSET VISIBLE TO 3

Complete • exact analysis of 658,008 hands • 40 cards • seed 5850701 • campaign plan complete; threat ordering and counter scores are guidance, not a guaranteed win rate.

OWNED INVENTORY

#	CARD	TYPE	ATK	C
1	Blue-eyes White Dragon	Dragon	3000	25
2	Mystical Elf	Spellcaster	800	20
3	Hitotsu-me Giant	Beast-Warrior	1200	10
4	Baby Dragon	Dragon	1200	70
5	Ryu-kishin	Fiend	1000	50
6	Feral Imp	Fiend	1300	14
7	Winged Dragon #1	Dragon	1400	12
8	Mushroom Man	Plant	800	60
9	Shadow Specter	Zombie	500	20
10	Blackland Fire Dragon	Dragon	1500	80
11	Sword Arm of Dragon	Dinosaur	1750	20
12	Swamp Battleguard	Warrior	1800	15

ATK ≥ 28000.00 %

ATK ≥ 25000.00 %

EXPECTED BEST1,734

DEAD HAND9.36 %

General campaign safety (33 opponents; final gauntlet excluded) • Best found; not a proof of global optimality.

Counter coverage score: 1,780 across 198 concrete threats. This is a heuristic, not a duel-win probability. Final-gauntlet tie-break score: 1,003.

Virtual purchase plan: spend 70 of 70 saved Star Chips; 0 remain. Nothing written to the save. Password-redemption history is not stored by the game; any names entered above were excluded from this plan.

DECKOUTCOMESLIMITSLOW VALUESTAR CHIPSTHREATS

CARD	PASSWOR	QTY	EACH	TOTAL	SPENT	LEFT	WHY THIS PURCHASE
Arlownay	14708569	1	45	45	45	25	Adds a legal copy used by the resulti
Ancient Jar	81492226	1	15	15	60	10	Adds a legal copy used by the resulti
Skull Servant	32274490	1	10	10	70	0	Adds a legal copy used by the resulti

Offline database ready • 722 cards • 25,146 resolved fusion pairs

1

Choose campaign scope, target duelist, profile, field, and types.

2

Enter or load owned quantities; visible rows match the calculation.

3

Saved Star Chips are optional and only create a virtual purchase plan.

4

Enter already redeemed password-card names so they are excluded.

5

Safety summary, exact hand metrics, purchases, threats, outcomes, limits, and exclusions explain the result.

YFM Fusion Companion - public Windows x64 guide

Page 9

Compact Live

A narrow right-side companion that leaves the central duel area visible

BEST LEGAL ROUTES

■

PI

5

L

RESULT	ATK	ROUTE
Twin-headed Th Dragon1	2800	1+2+? O> F1✓ O>al F1=F2?2
Pumpkin the King of Ghosts	1800	F(3)+23 C>O> F2? O>al>li F3?
Armored Zombie	1500	F(8)+44 O>C> — C>O>li

1

Result name and effective ATK.

3

F(1)-F(5) are monster positions; F(6)-F(10) are spell/trap positions.

5

PIN stays above the game; FULL restores the normal window.

2

Hand routes use plain slot numbers.

4

Two guardian chains and F# outcomes stay beneath the route; ? means unknown.

Compatibility, privacy, and troubleshooting

Compatibility boundary

- Manual adviser, analyzer, and optimizer work from the bundled offline catalog.
- Live decoding and automatic discovery are validated for NTSC-U / SLUS-01411 in RetroArch/SwanStation.
- Other regions, revisions, emulators, cores, and modified games are not claimed compatible; use manual tools.
- RetroAchievements Hardcore Mode works because the companion never uses save states.

Read-only and private by design

- No game input, memory writes, save edits, config edits, telemetry, updater, ads, or online API.
- Local settings store window bounds, compact/topmost choices, and the last selected save path.
- Diagnostics are created only on request and exclude gameplay/card values.
- No ROM, BIOS, emulator, memory-card save, or copyrighted card artwork is distributed.

When something is unavailable

- ERROR or DISCONNECTED: confirm Network Commands, port 55355, firewall scope, SwanStation, and that RetroArch was restarted.
- NOT IN DUEL: connectivity is healthy; enter a duel. Stale duel structures are intentionally hidden in story mode.
- Save looks old: save in-game, close content so the emulator flushes the file, then Refresh Saved Snapshot.
- Wrong region/core: Live Duel may be rejected safely, but every manual feature remains available.
- For support, use Export Diagnostics; review the text yourself before sharing it.